Explanatory Analysis

Winter Quarter 2026, Georgia Vasey

Explanatory Analysis

Now that I completed the exploratory phase (DAG → exploratory plots → correlation checks → distributions → response check), it is time to start finalizing the models. I first wanted to complete models for the change in soil C response variable.

Change in soil C

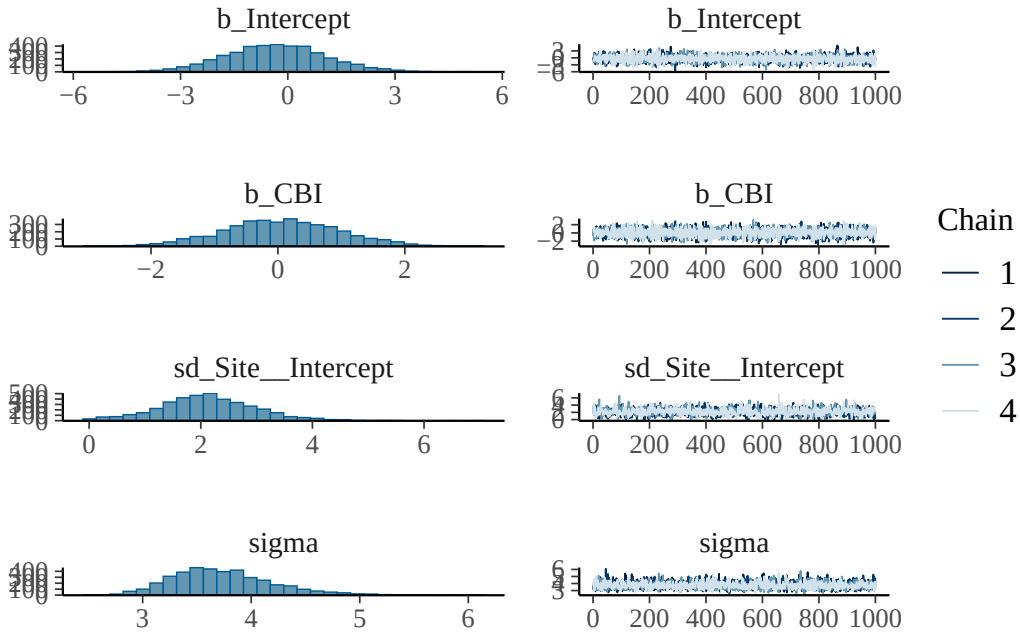
```
# Read in the data
analysis_carbon <- readRDS(here::here("data", "analysis_carbon.rds"))

# Load the libraries
library(tidyverse) # For data wrangling
library(brms) # For stats
library(bayesplot)
library(ggeffects) # for plotting model predictions

## Standardizing the predictors
 # Improves model convergence, coefficient interpretation, comparison of effect sizes
analysis_carbon <- analysis_carbon %>%
  mutate(
    precip_z = as.numeric(scale(`5y_precipitation`)),
    moisture_z = as.numeric(scale(days_since_precip)),
    slope_z = as.numeric(scale(slope)),
    clay_z = as.numeric(scale(clay)),
    RDM_z = as.numeric(scale(RDM_g_preburn))
  )
## Coefficients represent effect per 1 SD change
 # Predictors are on comparable scales
```

```r
 # Mixed models converge more easily

# Create dataset - Use the standardized predictors, but filtered for burn transects
burn_soilC <- analysis_carbon %>%
  filter(Treatment == "Burn") %>%
  select(
    Site,
    final_C_Mg_per_ha,
    CBI,
    precip_z,
    moisture_z,
    slope_z,
    clay_z
  ) %>%
  na.omit()

# Set the priors
priors_soilC <- c(
  prior(normal(0, 3), class = "Intercept"),
  prior(normal(0, 1), class = "b"),
  prior(exponential(1), class = "sigma"),
  prior(exponential(1), class = "sd")
)
## A normal(0, 1) prior on slopes says that there is an expectation that most effects are mo

# Model 0 - Reference Model
m0_soilC <- brm(data = burn_soilC, # Reduced dataframe with burn-only transects
                family = gaussian, # normal distribution
                # Model with only severity and site random intercept
                final_C_Mg_per_ha ~ CBI + (1 | Site),
                # Specifying the standard priors
                prior = priors_soilC,
                # Defaults from class
                iter = 2000, warmup = 1000, chains = 4, cores = 4,
                # Saving the model output
                file = "output/m.0_soilC")

# Check if the model is normal
summary(m0_soilC) #Rhat = 1
```

```
 Family: gaussian
  Links: mu = identity
```

```
Formula: final_C_Mg_per_ha ~ CBI + (1 | Site)
   Data: burn_soilC (Number of observations: 44)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~Site (Number of levels: 9)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     2.12      0.87     0.36     3.96 1.01     1227     1272

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept    -0.34      1.41    -3.04     2.49 1.00     3124     2475
CBI           0.06      0.91    -1.71     1.85 1.00     3965     2970

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     3.74      0.46     2.96     4.80 1.00     1870     1928

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
plot(m0_soilC) # Posterior distributions are normal, flat slope of chains, and chains are not
```




```
#print(m.0_soilC, digits = 3) - use when it is close to 0
```

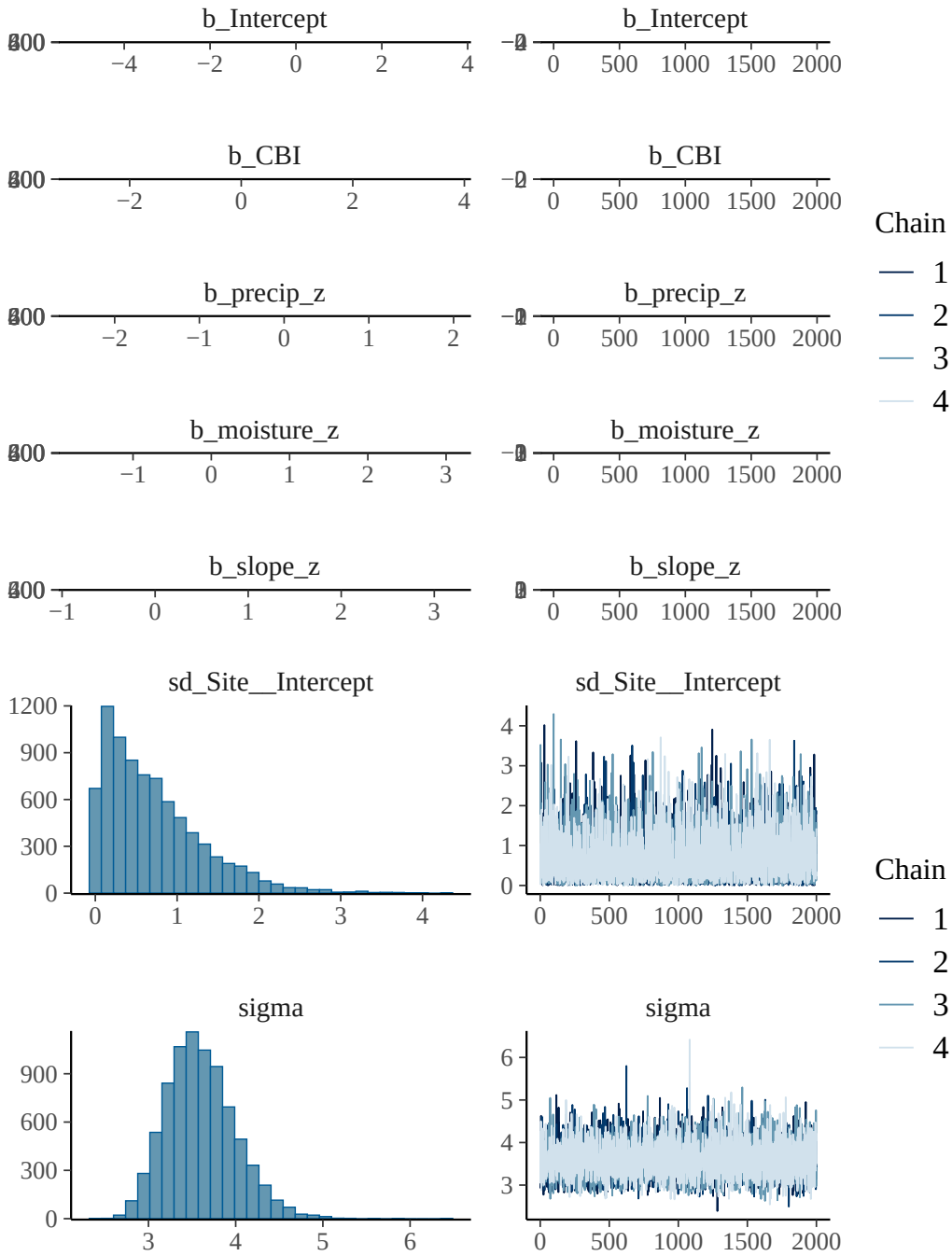
Now I will run the code for each of the other model selections. This is for Model 1 (strict DAG adjustment). From the results, slope is the only predictor that has an estimate that does not include 0. A 1 standard deviation increase in slope is associated with an estimated increase of 1.14 Mg C ha ¹ change in soil carbon.

```
# Model 1 - Strict DAG Model
m1_soilC <- brm(data = burn_soilC, # Reduced dataframe with burn-only transects
                family = gaussian, # normal distribution
                # Model with only severity and site random intercept
                final_C_Mg_per_ha ~ CBI + precip_z + moisture_z +
                  slope_z + (1 | Site),
                # Specifying the standard priors
                prior = priors_soilC,
                # Defaults from class
                iter = 4000, warmup = 2000, chains = 4, cores = 4,
                # Change how cautious Stan explores posterior (default is 0.8)
                control = list(adapt_delta = 0.99),
                # Saving the model output
                file = "output/m.1_soilC")
```

```r
# Check if the model is normal
summary(m1_soilC) #Rhat = 1
```

```
 Family: gaussian
  Links: mu = identity
Formula: final_C_Mg_per_ha ~ CBI + precip_z + moisture_z + slope_z + (1 | Site)
   Data: burn_soilC (Number of observations: 44)
  Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~Site (Number of levels: 9)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.71      0.60     0.02     2.17 1.00     3929     4562

Regression Coefficients:
           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept     -0.98      1.17    -3.25     1.36 1.00    10003     5374
CBI            0.49      0.86    -1.20     2.18 1.00    10968     5915
precip_z      -0.40      0.57    -1.53     0.74 1.00     7517     5960
moisture_z     0.88      0.62    -0.33     2.07 1.00     7383     5732
slope_z        1.14      0.56     0.04     2.23 1.00     6503     5467

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     3.60      0.40     2.92     4.46 1.00     9819     5871

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
# Predictor with CI that does not cross 0:
 # slope (1.14) with CI from 0.04 to 2.23
plot(m1_soilC)
```




```
# The posterior distributions are smooth with one clean peak, the four chains are overlapping
```

This is for Model 2 (replacing slope with clay). Now, the only predictor that has 95% CI without 0 is moisture_z (i.e., days since rainfall on burn date). A 1 SD increase in days since

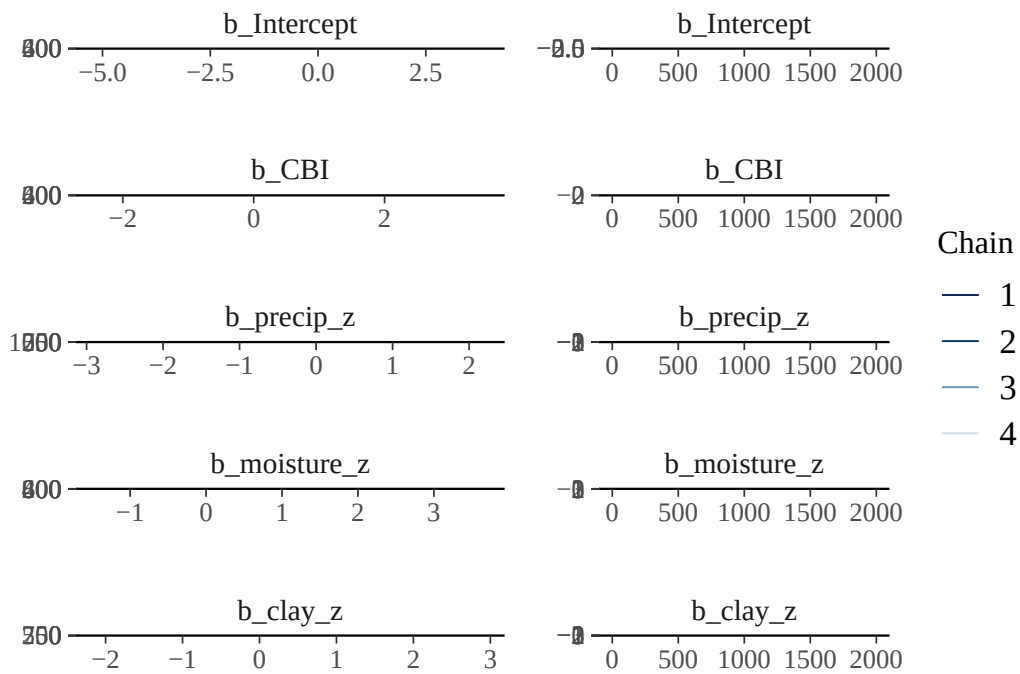
precipitation on burn date is 1.36 g greater change in soil carbon, with a 95% CI from 0.13 to 2.51.

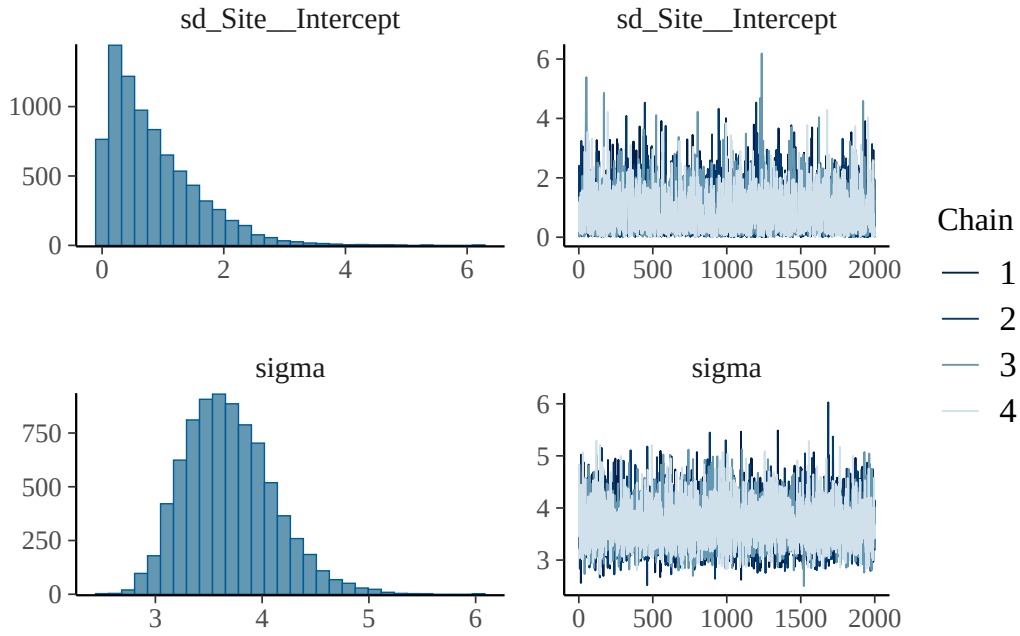
```r
# Model 2 - Replacing slope with clay
m2_soilC <- brm(data = burn_soilC, # Reduced dataframe with burn-only transects
                family = gaussian, # normal distribution
                # Model with only severity and site random intercept
                final_C_Mg_per_ha ~ CBI + precip_z + moisture_z +
                  clay_z + (1 | Site),
                # Specifying the standard priors
                prior = priors_soilC,
                # Defaults from class
                iter = 4000, warmup = 2000, chains = 4, cores = 4,
                # Change how cautious Stan explores posterior (default is 0.8)
                control = list(adapt_delta = 0.99),
                # Saving the model output
                file = "output/m.2_soilC")

# Check if the model is normal
summary(m2_soilC) #Rhat = 1
```

```
 Family: gaussian
  Links: mu = identity
Formula: final_C_Mg_per_ha ~ CBI + precip_z + moisture_z + clay_z + (1 | Site)
   Data: burn_soilC (Number of observations: 44)
  Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~Site (Number of levels: 9)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.84      0.71     0.03     2.57 1.00     2801     3983

Regression Coefficients:
           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept     -0.95      1.21    -3.32     1.46 1.00     7641     4986
CBI            0.53      0.89    -1.24     2.30 1.00     8437     5758
precip_z      -0.56      0.57    -1.69     0.55 1.00     7337     6003
moisture_z     1.36      0.61     0.13     2.51 1.00     7085     5233
clay_z         0.60      0.58    -0.57     1.72 1.00     6572     4865

Further Distributional Parameters:
```

```
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     3.69      0.42     2.99     4.61 1.00     8209     5580

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
# Predictor with CI that does not cross 0:
 # days_since_precip, called moisture_z, (1.36) with CI from 0.13 to 2.51
plot(m2_soilC)
```






```
# The posterior distributions are smooth with one clean peak, the four chains are overlapping
```

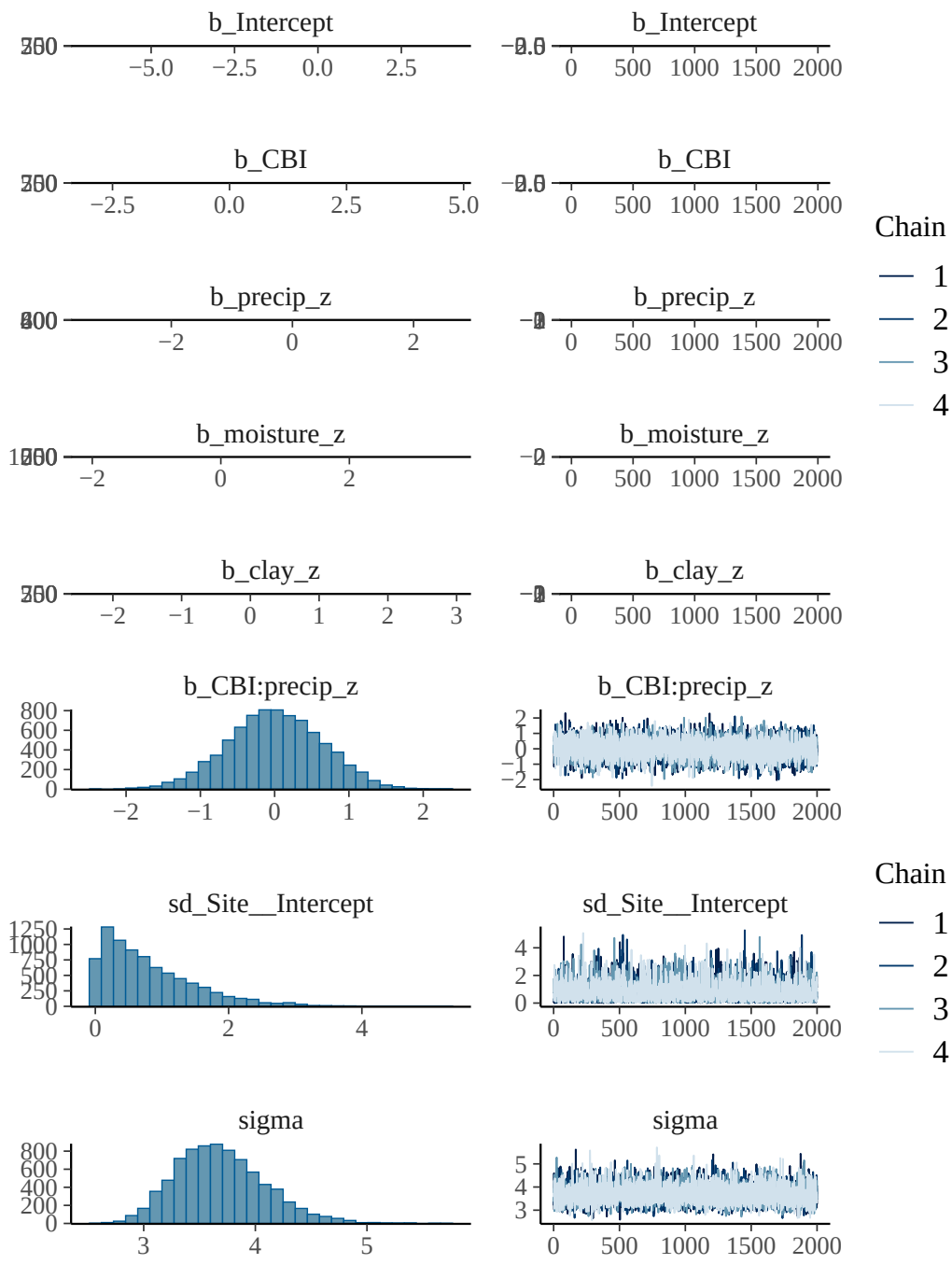
This is for model 3, which is testing whether precipitation affects burn severity. There appears to be no interaction between CBI and long-term precipitation, although days since rainfall is still a significant predictor. A 1 SD increase in days since precipitation on burn date is 1.36 g greater change in soil carbon, with a 95% CI from 0.12 to 2.48.

```
# Model 3 - Testing the precipitation interaction
 # Does the effect of burn severity depend on precipitation?
m3_soilC <- brm(data = burn_soilC, # Reduced dataframe with burn-only transects
                family = gaussian, # normal distribution
                # Model with only severity and site random intercept
                final_C_Mg_per_ha ~ CBI + precip_z + moisture_z +
                  clay_z + CBI:precip_z + (1 | Site),
                # Specifying the standard priors
                prior = priors_soilC,
                # Defaults from class
                iter = 4000, warmup = 2000, chains = 4, cores = 4,
                # Change how cautious Stan explores posterior (default is 0.8)
                control = list(adapt_delta = 0.99),
                # Saving the model output
                file = "output/m.3_soilC")
```

```r
# Check if the model is normal
summary(m3_soilC) #Rhat = 1
```

```
 Family: gaussian
  Links: mu = identity
Formula: final_C_Mg_per_ha ~ CBI + precip_z + moisture_z + clay_z + CBI:precip_z + (1 | Site)
   Data: burn_soilC (Number of observations: 44)
  Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~Site (Number of levels: 9)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.81      0.71     0.02     2.61 1.00     2500     2720

Regression Coefficients:
             Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept       -0.95      1.23    -3.36     1.49 1.00     7314     5102
CBI              0.54      0.88    -1.24     2.28 1.00     7605     6075
precip_z        -0.57      0.81    -2.16     1.00 1.00     5313     5597
moisture_z       1.36      0.60     0.12     2.48 1.00     5621     5095
clay_z           0.60      0.57    -0.58     1.69 1.00     6796     4689
CBI:precip_z     0.01      0.63    -1.24     1.23 1.00     5355     5938

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     3.69      0.40     3.00     4.58 1.00     7299     6168

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
 # Fire severity effects do NOT appear to depend on climate gradient.
plot(m3_soilC)
```




```
# The posterior distributions are smooth with one clean peak, the four chains are overlapping
```

This is for model 4 (Site as a fixed effect), with only days since precipitation being an important

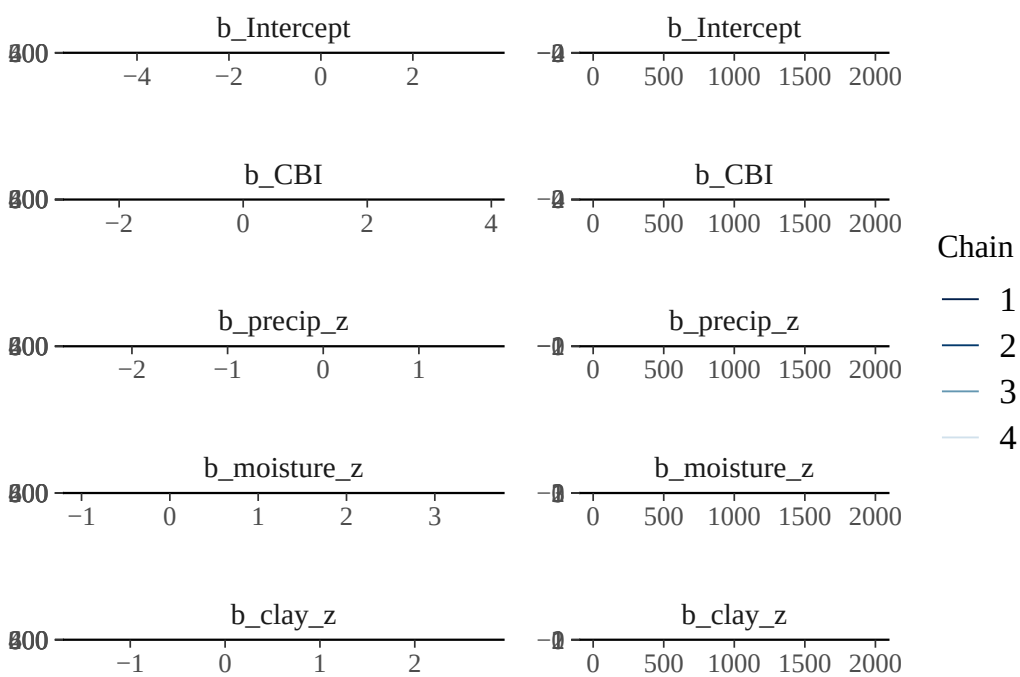
predictor. A 1 SD increase in days since precipitation on burn date is 1.36 g greater change in soil carbon, with a 95% CI from 0.17 to 2.51.

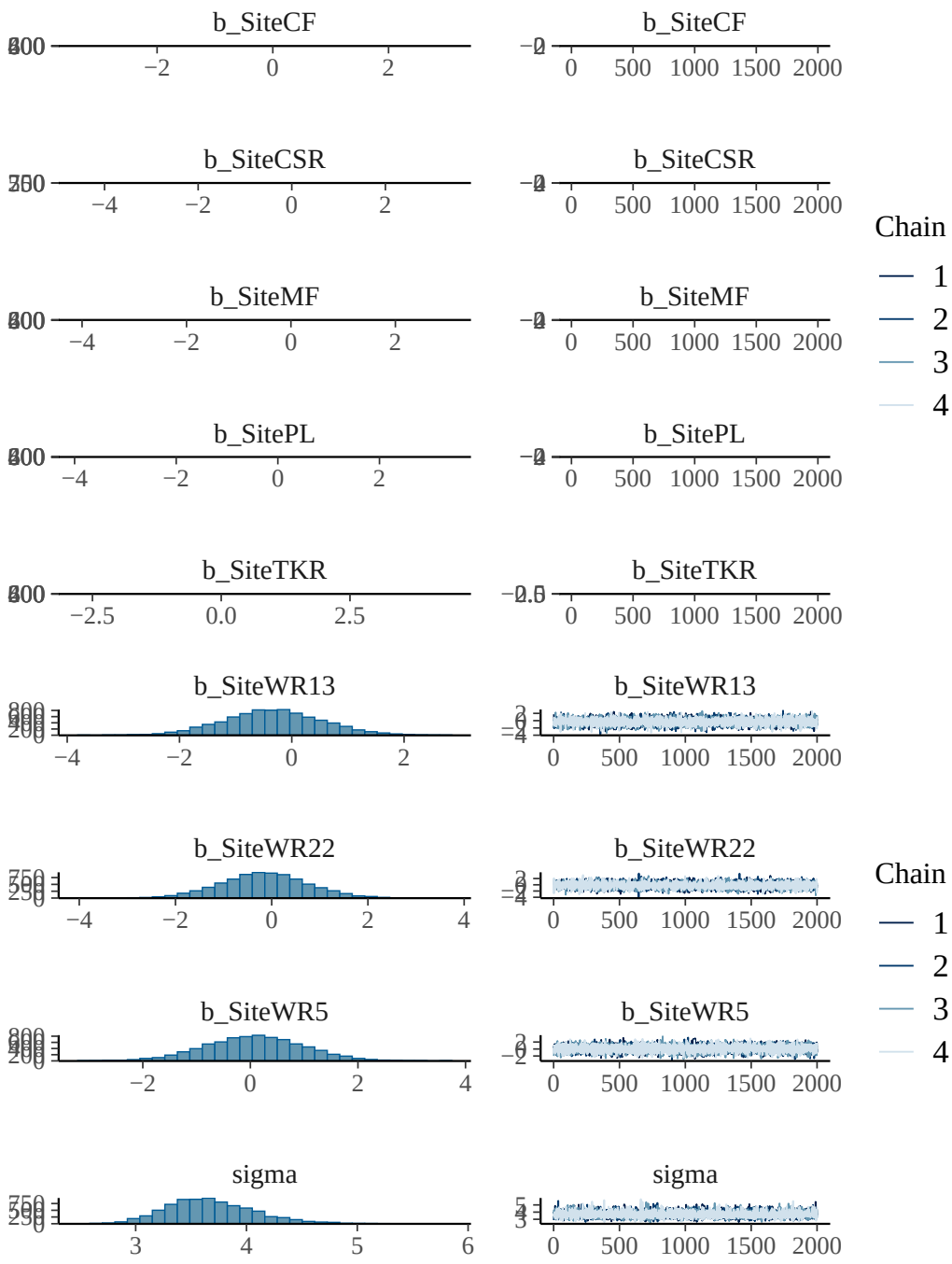
```r
# Need to create a new prior for the fixed-effects model that removes the sd prior
priors_soilC_fixed <- c(
  prior(normal(0, 2), class = "Intercept"),
  prior(normal(0, 1), class = "b"),
  prior(exponential(1), class = "sigma")
)

# Model 4 - Site as a fixed effect
m4_soilC <- brm(data = burn_soilC, # Reduced dataframe with burn-only transects
                family = gaussian, # normal distribution
                # Model with only severity and site random intercept
                final_C_Mg_per_ha ~ CBI + precip_z + moisture_z +
                  clay_z + Site,
                # Specifying the standard priors
                prior = priors_soilC_fixed,
                # Defaults from class
                iter = 4000, warmup = 2000, chains = 4, cores = 4,
                # Change how cautious Stan explores posterior (default is 0.8)
                control = list(adapt_delta = 0.99),
                # Saving the model output
                file = "output/m.4_soilC")

# Check if the model is normal
summary(m4_soilC) #Rhat = 1
```

```
 Family: gaussian
  Links: mu = identity
Formula: final_C_Mg_per_ha ~ CBI + precip_z + moisture_z + clay_z + Site
   Data: burn_soilC (Number of observations: 44)
  Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 8000

Regression Coefficients:
           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept     -0.92      1.19    -3.25     1.41 1.00     9634     6278
CBI            0.58      0.86    -1.15     2.27 1.00    10248     5884
precip_z      -0.51      0.56    -1.62     0.57 1.00     8747     5828
moisture_z     1.36      0.59     0.17     2.51 1.00     8883     6337
clay_z         0.56      0.56    -0.53     1.63 1.00     8864     6029
```

```
SiteCF        -0.25      0.85    -1.91      1.42 1.00    10948     6029
SiteCSR       -0.18      0.93    -2.00      1.61 1.00    10900     6784
SiteMF        -0.65      0.91    -2.40      1.15 1.00     9597     5949
SitePL         0.01      0.94    -1.82      1.87 1.00     9599     5741
SiteTKR        0.86      0.95    -0.96      2.75 1.00     9953     6225
SiteWR13      -0.33      0.87    -2.03      1.41 1.00    10210     6024
SiteWR22      -0.15      0.90    -1.91      1.63 1.00     9884     5932
SiteWR5        0.08      0.91    -1.73      1.86 1.00    10516     6157

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     3.68      0.40     2.99     4.57 1.00     8970     6067

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
 # Fire severity effects do NOT appear to depend on climate gradient.
plot(m4_soilC)
```






```
# The posterior distributions are smooth with one clean peak, the four chains are overlapping
```

Now, I want to compare the model fit using PSIS and WAIC. What I found is that Model 1 was the best fit model, with slope explaining variation slightly better than clay (PSIS for

Model 1 = 247.7 vs. Model 0 = 253.4, Model 2 = 249.9, Model 3 = 250.2, and Model 4 = 249.8). Since slope and days since precipitation are correlated (0.57), this is likely why days since precipitation (i.e., moisture_z) was not a predictive estimate in Model 1.

```
#PSIS
 # ΔwaicIC   2-3 which is very small
loo(m0_soilC) # 253.4
```

```
Computed from 4000 by 44 log-likelihood matrix.

         Estimate   SE
elpd_loo   -126.7  6.1
p_loo         8.6  2.2
looic       253.4 12.2
------
MCSE of elpd_loo is NA.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.4, 1.5]).

Pareto k diagnostic values:
                         Count Pct.    Min. ESS
(-Inf, 0.7]   (good)     43    97.7%   441
   (0.7, 1]   (bad)       1     2.3%   <NA>
   (1, Inf)   (very bad)  0     0.0%   <NA>
See help('pareto-k-diagnostic') for details.
```

```
loo(m1_soilC) # 247.7
```

```
Computed from 8000 by 44 log-likelihood matrix.

         Estimate   SE
elpd_loo   -123.8  5.9
p_loo         6.7  1.8
looic       247.7 11.8
------
MCSE of elpd_loo is 0.1.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.8, 1.5]).

All Pareto k estimates are good (k < 0.7).
See help('pareto-k-diagnostic') for details.
```

```r
 # According to waicIC, slope explains variation slightly better than clay.
 # This follows the DAG, has best predictive performance, mechanistically interpretable
loo(m2_soilC) # 249.9
```

```
Computed from 8000 by 44 log-likelihood matrix.

         Estimate   SE
elpd_loo   -124.9  6.1
p_loo         6.8  1.9
looic       249.9 12.2
------
MCSE of elpd_loo is 0.1.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.7, 1.6]).

All Pareto k estimates are good (k < 0.7).
See help('pareto-k-diagnostic') for details.
```

```r
loo(m3_soilC) # 250.2
```

```
Computed from 8000 by 44 log-likelihood matrix.

         Estimate   SE
elpd_loo   -125.1  6.1
p_loo         7.0  1.9
looic       250.2 12.3
------
MCSE of elpd_loo is 0.1.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.7, 1.5]).

All Pareto k estimates are good (k < 0.7).
See help('pareto-k-diagnostic') for details.
```

```r
loo(m4_soilC) # 249.8
```

```
Computed from 8000 by 44 log-likelihood matrix.

         Estimate   SE
```

```
elpd_loo   -124.9  6.1
p_loo         6.7  1.9
looic       249.8 12.2
------
MCSE of elpd_loo is 0.1.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.6, 1.7]).

All Pareto k estimates are good (k < 0.7).
See help('pareto-k-diagnostic') for details.
```

```
 ## slope and moisture are correlated (0.57)
 ## soil carbon responses depend on something related to dryness / landscape position
  # Potentially slope-driven drainage, burn dryness, fuel conditions
 ## Soil carbon responses to fire were highly variable and not strongly explained by fire se

#WAIC
waic(m0_soilC) # 252.9
```

```
Computed from 4000 by 44 log-likelihood matrix.

          Estimate   SE
elpd_waic   -126.4  6.0
p_waic         8.4  2.1
waic         252.9 12.0

5 (11.4%) p_waic estimates greater than 0.4. We recommend trying loo instead.
```

```
waic(m1_soilC) # 247.1 -- still the best model
```

```
Computed from 8000 by 44 log-likelihood matrix.

          Estimate   SE
elpd_waic   -123.6  5.8
p_waic         6.4  1.7
waic         247.1 11.6

4 (9.1%) p_waic estimates greater than 0.4. We recommend trying loo instead.
```

```r
waic(m2_soilC) # 249.4
```

```
Computed from 8000 by 44 log-likelihood matrix.

          Estimate   SE
elpd_waic   -124.7  6.0
p_waic         6.6  1.8
waic         249.4 12.1

4 (9.1%) p_waic estimates greater than 0.4. We recommend trying loo instead.
```

```r
waic(m3_soilC) # 249.7
```

```
Computed from 8000 by 44 log-likelihood matrix.

          Estimate   SE
elpd_waic   -124.8  6.0
p_waic         6.7  1.8
waic         249.7 12.1

4 (9.1%) p_waic estimates greater than 0.4. We recommend trying loo instead.
```

```r
waic(m4_soilC) # 249.3
```

```
Computed from 8000 by 44 log-likelihood matrix.

          Estimate   SE
elpd_waic   -124.7  6.0
p_waic         6.5  1.8
waic         249.3 12.1

4 (9.1%) p_waic estimates greater than 0.4. We recommend trying loo instead.
```

```r
# Key Takeaway
# Fire severity does not strongly control soil carbon responses in coastal prairie burns.
```

Lastly, I want to be able to explain what "1 standard deviation" of change is for different predictors, since the predictors were scaled. Predictors were standardized so that model coefficients represent the effect of a one standard deviation change in each variable. For slope in soil carbon data (Model 1), one standard deviation corresponds to roughly 3.7 slope units, meaning steeper slopes were associated with approximately 1.14 Mg C ha ^{1} greater soil carbon change after fire.

```
## Back-transform from 1 standard deviation
 # From model 1, the estimate for slope_z was 1.14
 # A 1 standard deviation increase in slope is associated with an estimated +1.14 Mg C ha ¹ c

# Calculate SD and mean of original data
mean(analysis_carbon$slope, na.rm = TRUE) #4.471264
```

```
[1] 4.471264
```

```
sd(analysis_carbon$slope, na.rm = TRUE) # 3.694242 (slope units)
```

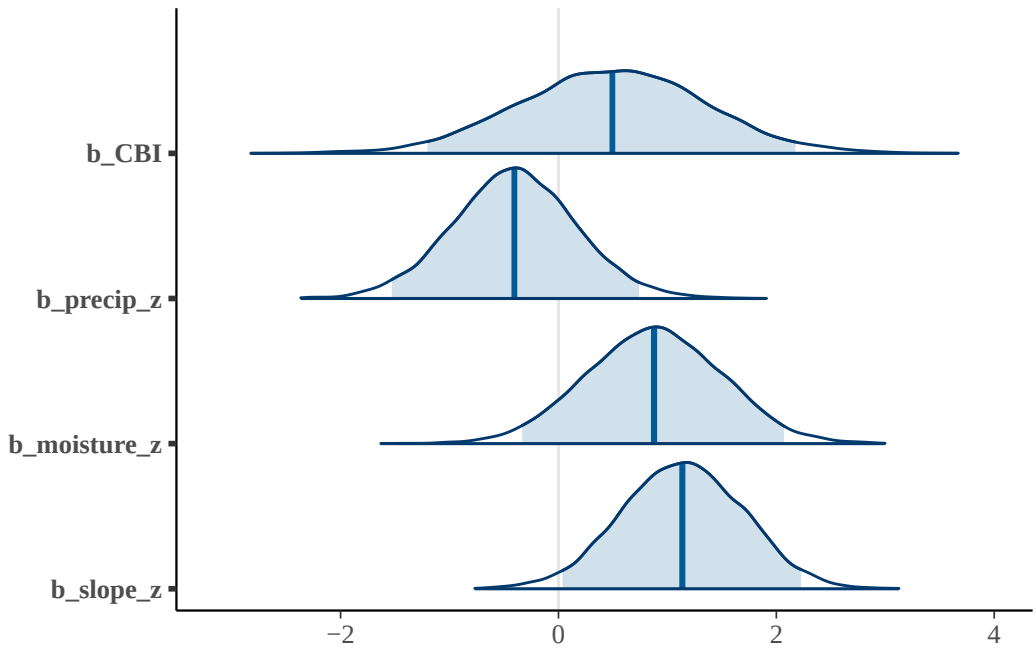
```
[1] 3.694242
```

```
#z = (x- mean)/SD
 # If I knew the slope of the transect (x), I would subtract the mean and divide by the slope

# Predictors were standardized so that model coefficients represent the effect of a one stand

## Visualization:

#Low slope      Mean slope      High slope
#   |--------------|--------------|
# 0.8             4.5             8.2
#(-1 SD)          (0)          (+1 SD)
```

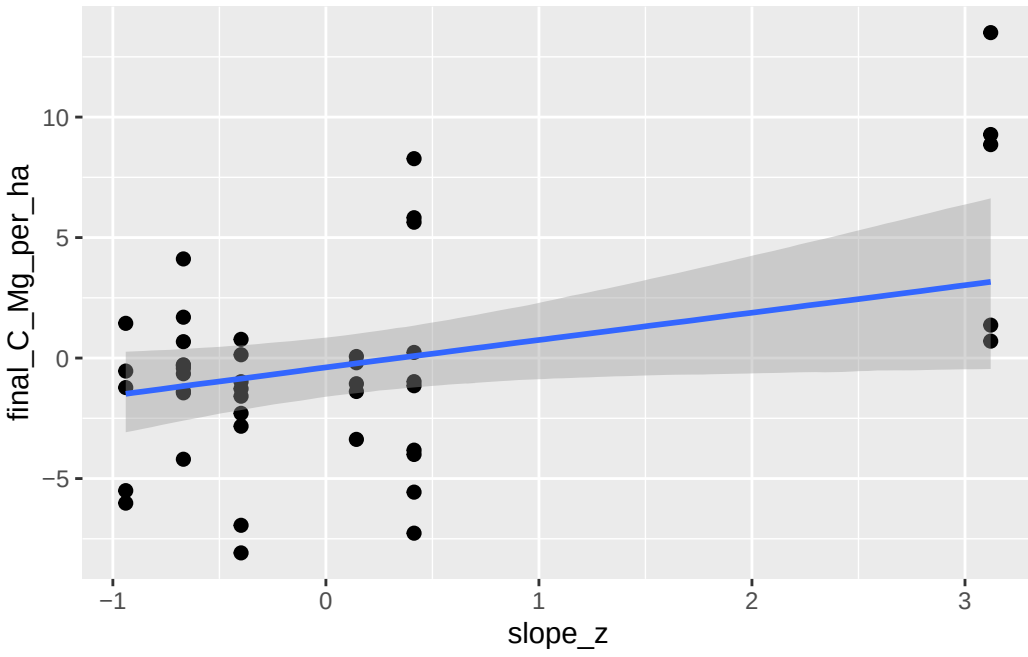
Time to visualize the model results. The first plot shows that soil carbon repsonses to prescribed fire were highly variable, and were more strongly associated with landscape position than with burn severity. The second plot shows that with steeper slopes, there is larger soil C change.

```
# Load library
library(bayesplot)
```

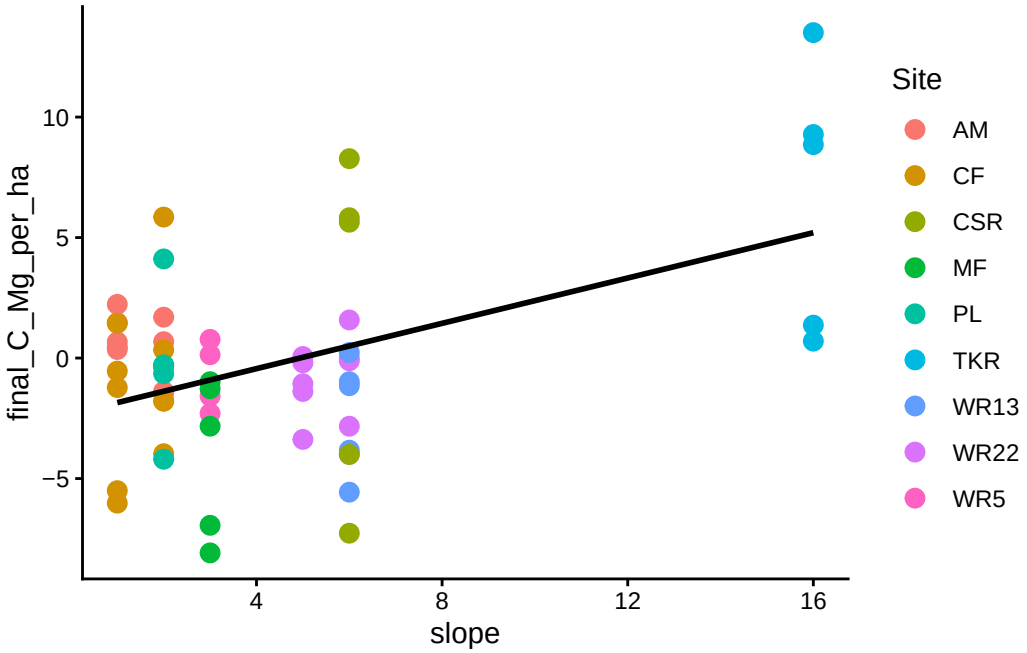
```r
# Coefficient (Posterior) Plot
coef_plot <-mcmc_areas(
  as.matrix(m1_soilC),
  pars = c("b_CBI","b_precip_z","b_moisture_z","b_slope_z"),
  prob = 0.95
)
print(coef_plot)
```




```r
# CBI = around 0 (Fire severity → weak effect)
# Long-term precipitation = around 0 (Climate → weak effect)
# Moisture = slightly positive
# Slope = positive effect (Landscape position (slope) → strongest signal)

# Marginal effects plot (predicted relationship)
ce_slope_soil <- conditional_effects(m1_soilC, effects = "slope_z")
plot(ce_slope_soil, points = TRUE)
```




```
# Ideally need to label axis is real slope units
## Steeper slopes → larger soil C change

# Raw data scatter plot (colored by Site)
ggplot(analysis_carbon, aes(slope, final_C_Mg_per_ha, color = Site)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  theme_classic()
```



Root biomass Post-burn

I only came up with one model for root biomass after looking at the DAG and exploratory plots. It is clear that Treatment (Burn vs. Control) is the most important predictor since in the reference model without covariates (Model 0), Treatment still has high predictive power. The estimate for TreatmentControl is -0.41 with a 95% CI from -0.80 to -0.01.

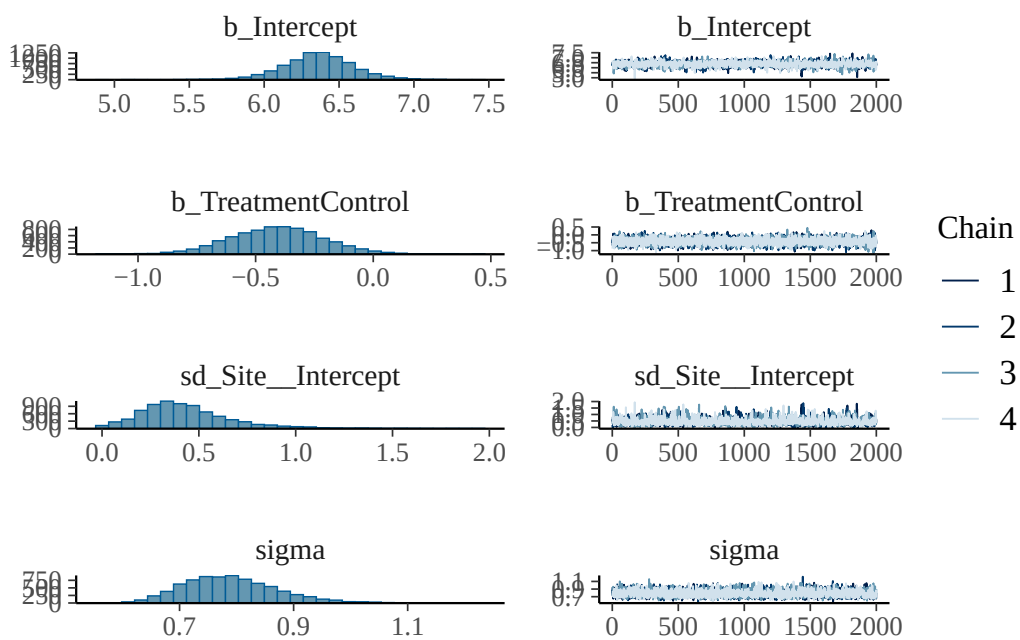
```
## Final model based on DAG and exploratory plots
# Use the standardized predictors, but filtered for Year = 2024
roots_new <- analysis_carbon %>%
  filter(Year == "2024") %>%
  select(
    Site,
    Roots_g_per_m2,
    Treatment,
    RDM_z,
    precip_z,
    clay_z,
    slope_z
  ) %>%
  na.omit()
```

```r
# Set the priors
priors_roots <- c(
  prior(normal(6, 2), class = "Intercept"),
  prior(normal(0, 0.5), class = "b"),
  prior(exponential(1), class = "sd"),
  prior(exponential(1), class = "sigma")
)

# Model 0 - Reference Model
m0_roots <- brm(data = roots_new, # Year == 2024 & scaled predictors
                # Estimating the log root biomass
                family = lognormal(), # log-normal distribution
                # Model with only severity and site random intercept
                Roots_g_per_m2 ~ Treatment + (1 | Site),
                # Specifying the standard priors
                prior = priors_roots,
                # Defaults from class
                iter = 4000, warmup = 2000, chains = 4, cores = 4,
                # Change how cautious Stan explores posterior (default is 0.8)
                control = list(adapt_delta = 0.99),
                # Saving the model output
                file = "output/m.0_roots")

# Check if the model is normal
summary(m0_roots) #Rhat = 1
```

```
 Family: lognormal
  Links: mu = identity
Formula: Roots_g_per_m2 ~ Treatment + (1 | Site)
   Data: roots_new (Number of observations: 57)
  Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~Site (Number of levels: 6)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.43      0.24     0.05     1.03 1.00     1688     1597

Regression Coefficients:
                 Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept            6.36      0.24     5.88     6.85 1.00     2998     3477
TreatmentControl    -0.41      0.20    -0.80    -0.01 1.00     6078     5278
```

```
Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.79      0.08     0.65     0.97 1.00     4451     4847

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
plot(m0_roots)
```




```
# The posterior distributions are smooth with one clean peak, the four chains are overlapping
```

Now, I want to compute the model with the environmental covariates. You can see that none of the environmental covariates have 95% CI that do not include zero (i.e., they are weak predictors). This compares with TreatmentControl which has an estimate of -0.47 and a 95% CI from -0.88 to -0.07. When you back-transform this value (e^0.47), burned transects are predicted to have about 60% higher root biomass than control transects.

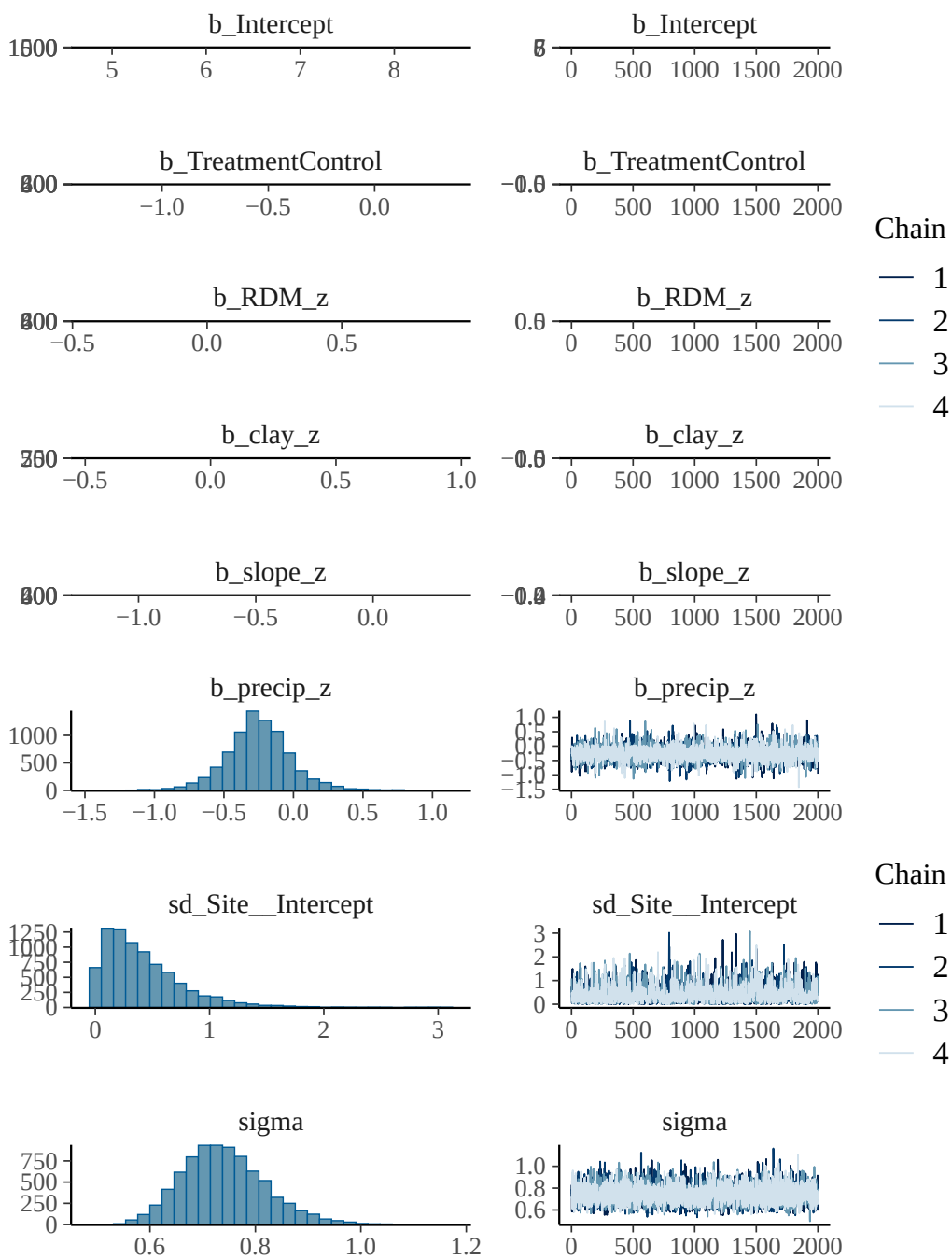
```
m1_roots <- brm(data = roots_new, # Year == 2024 & scaled predictors
                # Estimating the log root biomass
                family = lognormal(), # log-normal distribution
```

```r
                 # Model with only severity and site random intercept
                 Roots_g_per_m2 ~ Treatment + RDM_z + clay_z + slope_z +
                   precip_z + (1 | Site),
                 # Specifying the standard priors
                 prior = priors_roots,
                 # Defaults from class
                 iter = 4000, warmup = 2000, chains = 4, cores = 4,
                 # Change how cautious Stan explores posterior (default is 0.8)
                 control = list(adapt_delta = 0.99),
                 # Saving the model output
                 file = "output/m.1_roots")

# Check if the model is normal
summary(m1_roots) #Rhat = 1
```

```
 Family: lognormal
  Links: mu = identity
Formula: Roots_g_per_m2 ~ Treatment + RDM_z + clay_z + slope_z + precip_z + (1 | Site)
   Data: roots_new (Number of observations: 57)
  Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~Site (Number of levels: 6)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.42      0.35     0.02     1.30 1.00     1208     2688

Regression Coefficients:
                 Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept            6.36      0.29     5.82     7.01 1.00     3806     2749
TreatmentControl    -0.47      0.21    -0.88    -0.07 1.00     4163     5027
RDM_z                0.24      0.16    -0.08     0.54 1.00     4036     4918
clay_z               0.26      0.17    -0.09     0.59 1.00     6111     5181
slope_z             -0.32      0.22    -0.80     0.05 1.00     1786     4464
precip_z            -0.25      0.24    -0.72     0.24 1.00     4484     3719

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.74      0.08     0.60     0.92 1.00     4719     5362

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
```

```
scale reduction factor on split chains (at convergence, Rhat = 1).
```

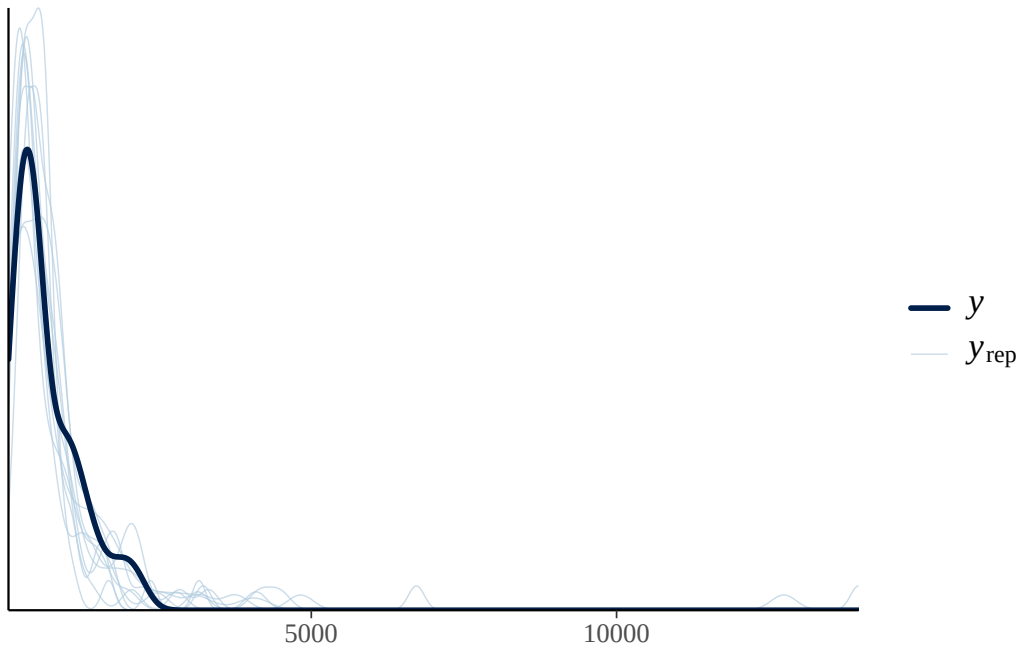
```
plot (m1_roots)
```



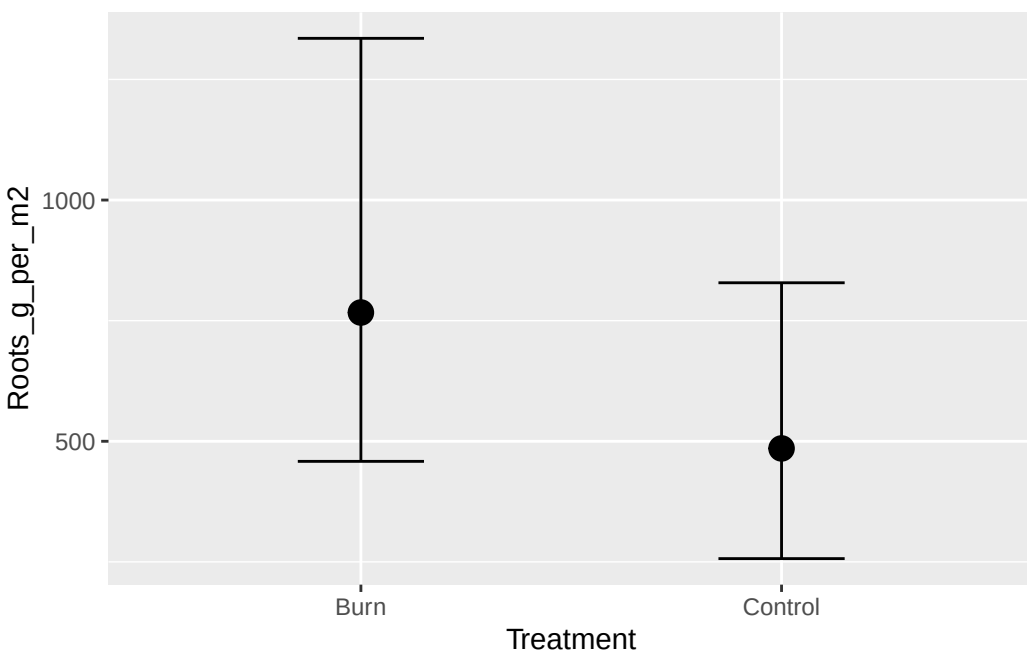
```r
#TreatmentControl estimate is 0.47 with a 95% CI from -0.07 to -0.88 ; everything else crosse
# Mean effect (converting from log root biomass)
exp(0.47) #1.599994 ~ 1.6
```

```
[1] 1.599994
```

```r
 # Control plots have 1.60 times less the root biomass of burn plots
 # Burned plots have 1.60 times the root biomass of control plots

# Rule for log models
# Whenever there is a coefficient  from a log model:
  # Percent change=(e^ -1)×100

# Final increase (back-transformed):
# Increase relative to control = (1.60-1) x 100 = 60%
# Burned transects are predicted to have about 60% higher root biomass than control transects

# Sd(Intercept) for Site = 0.42
 # Indicates substantial among-site variability in root biomass, even after controlling for p
# sigma = 0.74 - describes the variation among transects within sites

# Other diagnostic tests:
pp_check(m1_roots) # Simulated distributions are close to observed data (dark blue line)
```




```
conditional_effects(m1_roots, effects = "Treatment")
```



I finish with evaluating which model has the best fit. Even though the environmental covariates

were weak predictors, the PSIS for Model 1 was 837.7, which was lower looic value than the reference model of 842.5.

```
# PSIS
loo(m0_roots) #842.5
```

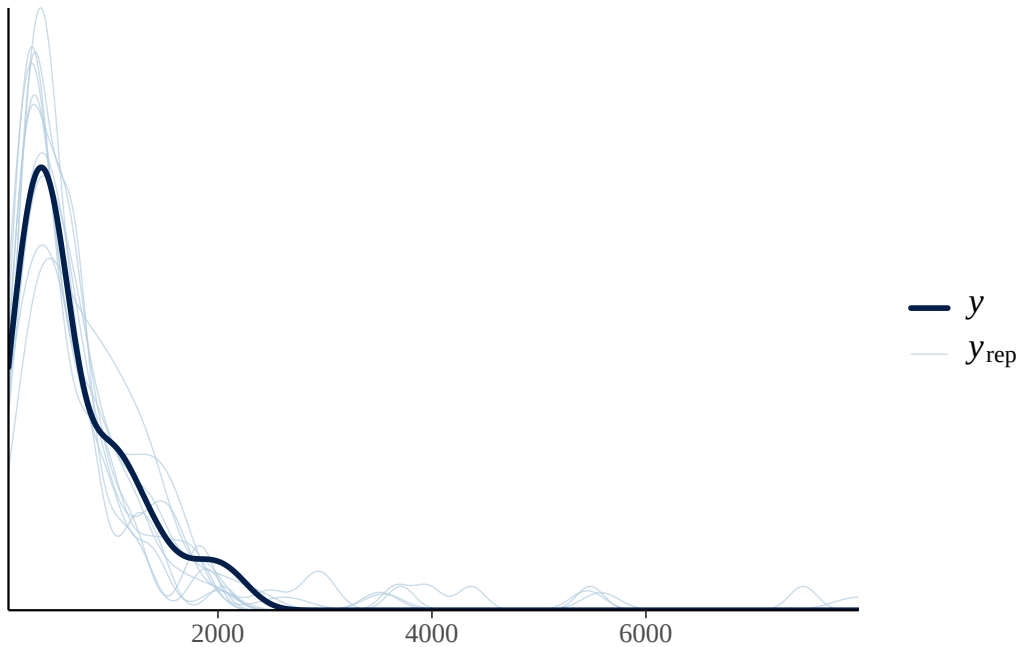
```
Computed from 8000 by 57 log-likelihood matrix.

         Estimate   SE
elpd_loo   -421.3  7.3
p_loo         6.4  1.3
looic       842.5 14.5
------
MCSE of elpd_loo is 0.1.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.4, 1.1]).

All Pareto k estimates are good (k < 0.7).
See help('pareto-k-diagnostic') for details.
```

```
loo(m1_roots) #837.7 --> better model with covariates
```

```
Computed from 8000 by 57 log-likelihood matrix.

         Estimate   SE
elpd_loo   -418.8  7.2
p_loo         8.3  1.7
looic       837.7 14.4
------
MCSE of elpd_loo is 0.1.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.4, 1.3]).

All Pareto k estimates are good (k < 0.7).
See help('pareto-k-diagnostic') for details.
```

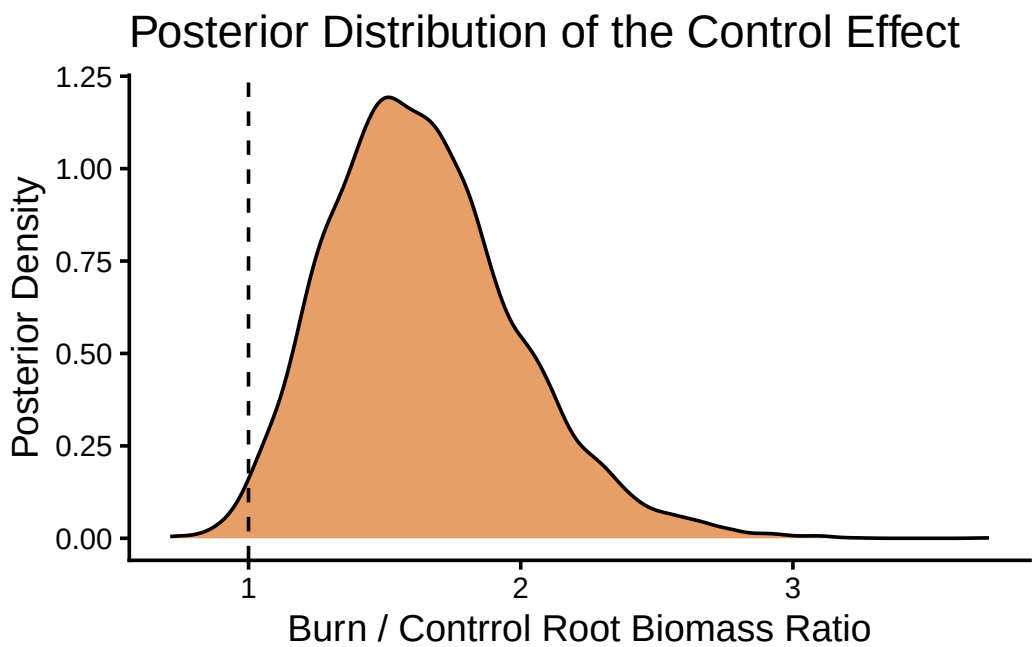
Lastly, I want to create visualizations from the root biomass model (m1_roots).

```
# Does the model reproduce the overall distribution of rootbiomass?
pp_check(m1_roots)
```




```
# Simulated distributions (lighter lines) are close to observed data (dark line)
 # Posterior predictive checks indicated that the lognormal mixed model adequately reproduce


# New library
library(tidybayes)

# Posterior distribution of the Burn effect
 # Need to add negative to the exp(-b_TreatmentControl)
control_draws_roots <- m1_roots %>%
  spread_draws(b_TreatmentControl) %>%
  mutate(effect_ratio = exp(-b_TreatmentControl))

# Plot the posterior probability for burn vs. control in root biomass
post_prob_roots <- ggplot(control_draws_roots, aes(effect_ratio)) +
  geom_density(fill = "#d95f02", alpha = 0.6) +
  geom_vline(xintercept = 1, linetype = "dashed") +
  theme_classic(base_size = 14) +
  labs(
    x = "Burn / Contrrol Root Biomass Ratio",
    y = "Posterior Density",
    title = "Posterior Distribution of the Control Effect"
  )
```

```r
print(post_prob_roots)
```



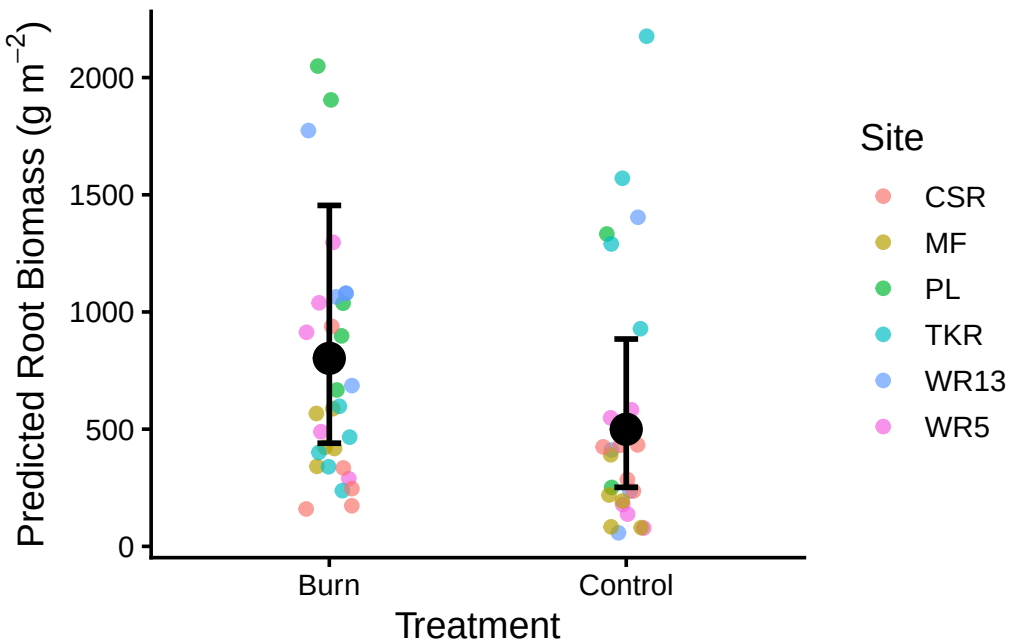

```r
## 1.0 line - no treatment effect
 # density to the right of 1 = higher roots in Burn plots
 # Posterior will mostly lie above 1, which visually shows the burn effect

# Calculate posterior probability
control_draws_roots %>%
  summarise(prob_burn_higher = mean(effect_ratio > 1))
```

```
# A tibble: 1 x 1
  prob_burn_higher
             <dbl>
1             0.99
```

```r
# 0.99
# There is about a 99% posterior probability that burned transects have greater root biomass

# Predicted root biomass by treatment (with raw data)
# Create a prediction dataset
 # Set predictors to 0 since they are standardized (mean values).
```

```r
newdat_roots <- data.frame(
  Treatment = c("Control", "Burn"),
  RDM_z = 0,
  clay_z = 0,
  slope_z = 0,
  precip_z = 0,
  Site = NA
)

# Get posterior predictions from that dataset
pred_draws_roots <- add_epred_draws(
  m1_roots,
  newdata = newdat_roots,
  re_formula = NA
)

# Summarize predictions
pred_summary_roots <- pred_draws_roots %>%
  group_by(Treatment) %>%
  summarise(
    mean = mean(.epred),
    lower = quantile(.epred, 0.025),
    upper = quantile(.epred, 0.975)
  )

# Plot predictions and raw data
roots_pred_raw <-
ggplot() +
  geom_jitter(
    data = roots_new,
    aes(Treatment, Roots_g_per_m2, color = Site),
    width = 0.08,
    alpha = 0.7
  ) +
  geom_point(
    data = pred_summary_roots,
    aes(Treatment, mean),
    size = 5,
    color = "black"
  ) +
  geom_errorbar(
    data = pred_summary_roots,
```

```
    aes(Treatment, ymin = lower, ymax = upper),
    width = 0.08,
    color = "black",
    linewidth = 1
  ) +
  theme_classic(base_size = 14) +
  labs(
    y = expression("Predicted Root Biomass (g m"^{-2}*")"),
    x = "Treatment"
  )
plot(roots_pred_raw)
```




```
# Interpretation:
  # colored points = individual transects
  # black points = posterior mean prediction
  # error bars = 95% credible interval
```

Change in aboveground biomass

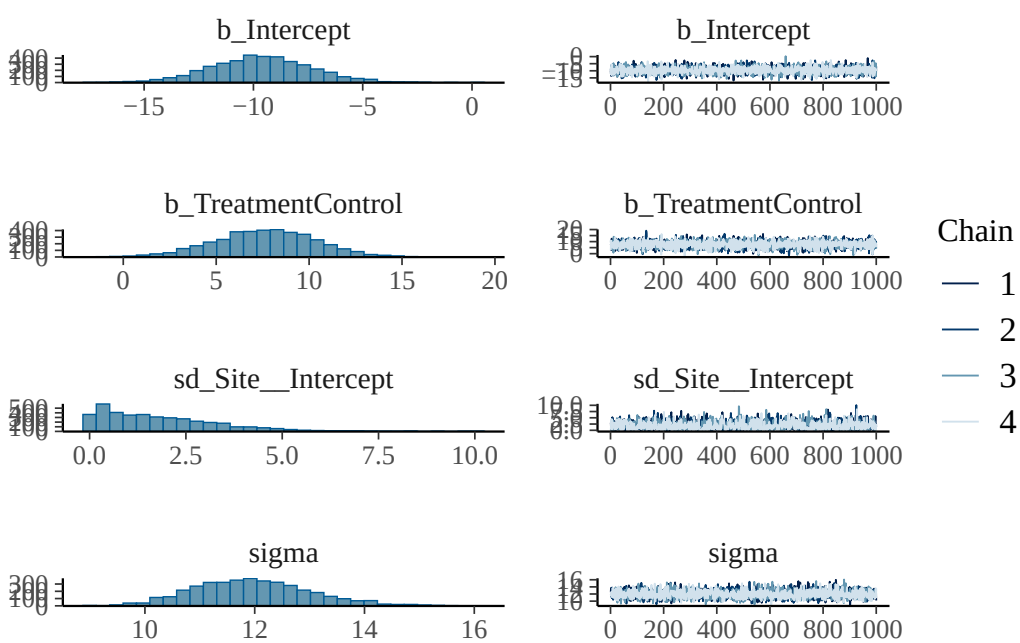
From my DAG and exploratory plots, I have two models that I want to run. I'm again using scaled predictors.

```r
# Use the standardized predictors, but filtered for burn transects
RDM_delta <- analysis_carbon %>%
  filter(Year == "2024") %>%
  select(
    Site,
    final_RDM,
    Treatment,
    RDM_z,
    precip_z,
    clay_z
  ) %>%
  na.omit()

# Set the priors
priors_RDM <- c(
  prior(normal(0, 20), class = Intercept),
  prior(normal(0, 5), class = "b"),
  prior(exponential(1), class = "sigma"),
  prior(exponential(1), class = "sd")
)
## A normal(0, 1) prior on slopes says that there is an expectation that most effects are mo

# Model 0 - Reference Model
m0_RDM <- brm(data = RDM_delta, # Reduced dataframe with burn-only transects
                family = gaussian, # normal distribution
                # Model with only severity and site random intercept
                final_RDM ~ Treatment + (1 | Site),
                # Specifying the standard priors
                prior = priors_RDM,
                # Defaults from class
                iter = 2000, warmup = 1000, chains = 4, cores = 4,
                # Change how cautious Stan explores posterior (default is 0.8)
                control = list(adapt_delta = 0.95),
                # Saving the model output
                file = "output/m.0_RDM")

# Check if the model is normal
summary(m0_RDM) #Rhat = 1
```

```
 Family: gaussian
  Links: mu = identity
Formula: final_RDM ~ Treatment + (1 | Site)
```

```
   Data: RDM_delta (Number of observations: 56)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~Site (Number of levels: 6)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     1.80      1.48     0.05     5.26 1.00     1554     1618

Regression Coefficients:
                 Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept           -9.65      2.29   -14.15    -5.03 1.00     3335     2851
TreatmentControl     7.64      2.76     2.11    12.84 1.00     3755     2918

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma    11.96      1.06    10.09    14.19 1.00     3508     2621

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
plot(m0_RDM)
```



```
# The posterior distributions are smooth with one clean peak, the four chains are overlapping
```

Now I will run the code for each of the other model selections.

This is for Model 1, which found burned transects (Intercept) experienced 13.17 g reduction in aboveground biomass. Similarly, control transects are 8.17 higher in the change in aboveground biomass (e.g., burn transects have larger biomass losses, whereas control has smaller losses) with 95% CI is 3.49 to 12.78.

Sites with higher pre-burn biomasss experienced greater reductions in biomass following fire, consistent with greater fuel loads resulting in greater biomass consumption. From the model, a 1 SD increase in pre-burn biomass leads to 7.37 g loss in biomass (95% CI is from -10.80 to -4.04). In addition, sites with higher clay content exhibited smaller declines in aboveground biomass, suggesting that soil texture influences vegetation recovery or productivity. From the model, 1 SD increase in % Clay leads to 4.03 g more change in abovegeground biomass (95% CI is from 1.16 to 6.72).

The site random effect sd(Intercept) is small at 0.88, meaning there is relatively little unexplained variation among sites (predictors capture most of the site differences), but that sigma = 9.95, meaning approximately 10g variation among transects is unexplained by the model.

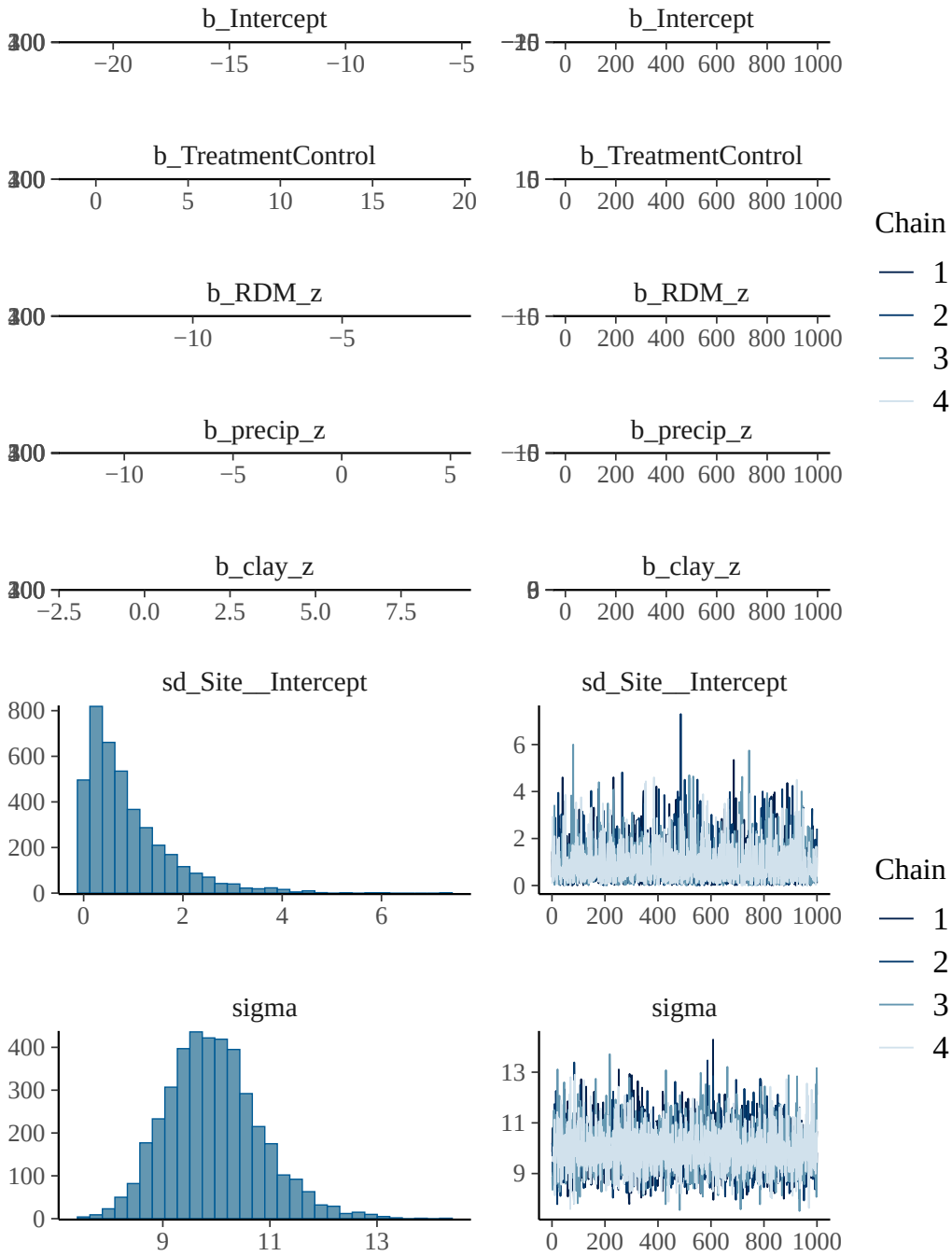
The fuel load effect (-7.37) is nearly as large as the treatment effect (8.17), meaning fuel load fire efect. **Fire impact depends strongly on how much biomass was present before the burn.**

```
m1_RDM <- brm(data = RDM_delta, # Reduced dataframe with burn-only transects
                family = gaussian, # normal distribution
                # Model with only severity and site random intercept
                final_RDM ~ Treatment + RDM_z + precip_z + clay_z + (1 | Site),
                # Specifying the standard priors
                prior = priors_RDM,
                # Defaults from class
                iter = 2000, warmup = 1000, chains = 4, cores = 4,
                # Change how cautious Stan explores posterior (default is 0.8)
                control = list(adapt_delta = 0.95),
                # Saving the model output
                file = "output/m.1_RDM")

# Check if the model is normal
summary(m1_RDM) #Rhat = 1
```

```
 Family: gaussian
  Links: mu = identity
```

```
Formula: final_RDM ~ Treatment + RDM_z + precip_z + clay_z + (1 | Site)
   Data: RDM_delta (Number of observations: 56)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~Site (Number of levels: 6)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.88      0.84     0.02     3.15 1.00     2780     2048

Regression Coefficients:
                 Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept          -13.17      2.13   -17.45    -8.90 1.00     5736     2984
TreatmentControl     8.17      2.45     3.49    12.78 1.00     6069     3136
RDM_z               -7.37      1.69   -10.80    -4.04 1.00     4373     3038
precip_z            -2.54      1.73    -5.99     0.76 1.00     4357     2857
clay_z               4.03      1.40     1.16     6.72 1.00     4544     2721

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     9.95      0.88     8.39    11.87 1.00     4755     2976

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
# The intercept (-13.17) with 95% CI from -17.45 to -8.90 (does not include 0)represents burn
 # Burned transects experienced an average 13 g reduction in RDM.
# Control transects have ~8 g higher ΔRDM than burned transects ; 95% CI = 3.49 to 12.78
# A 1 SD increase in pre-burn biomass leads to ~7 g greater biomass loss.
plot(m1_RDM)
```




```
# The posterior distributions are smooth with one clean peak, the four chains are overlapping
```

This is for Model 2, which looks at the interaction of Pre-burn RDM with Treatment (Burn vs. Control) or the fuel-load relationship. These results have similar estimates, but the inter-

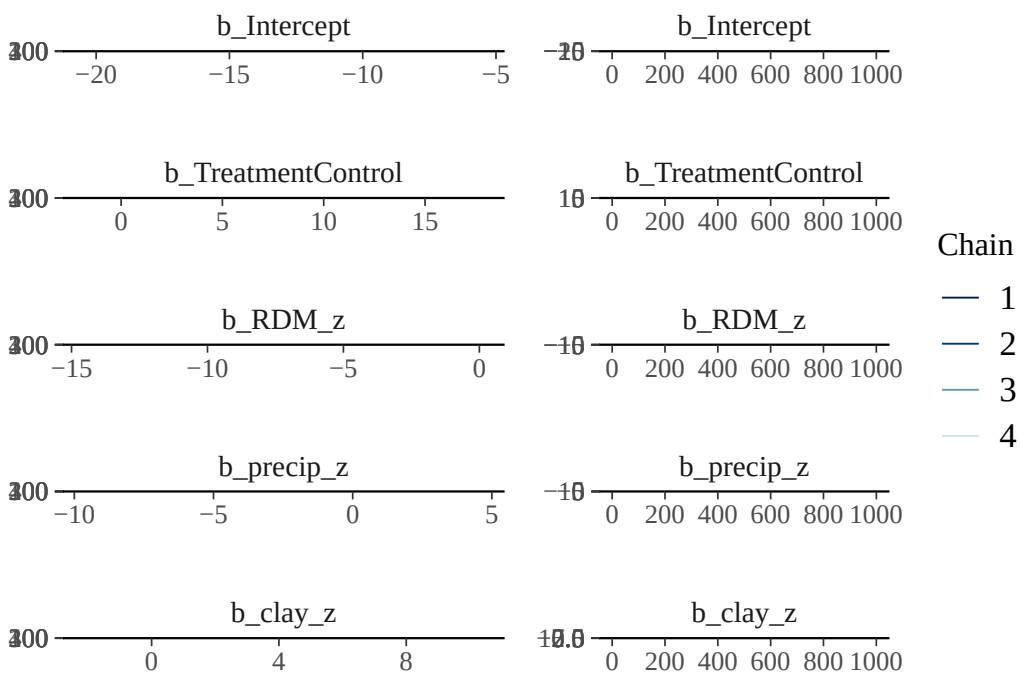
action estimate has a 95% CI that crosses 0, suggesting that there is no strong evidence that the fuel-load relationship differs between the burn and control.

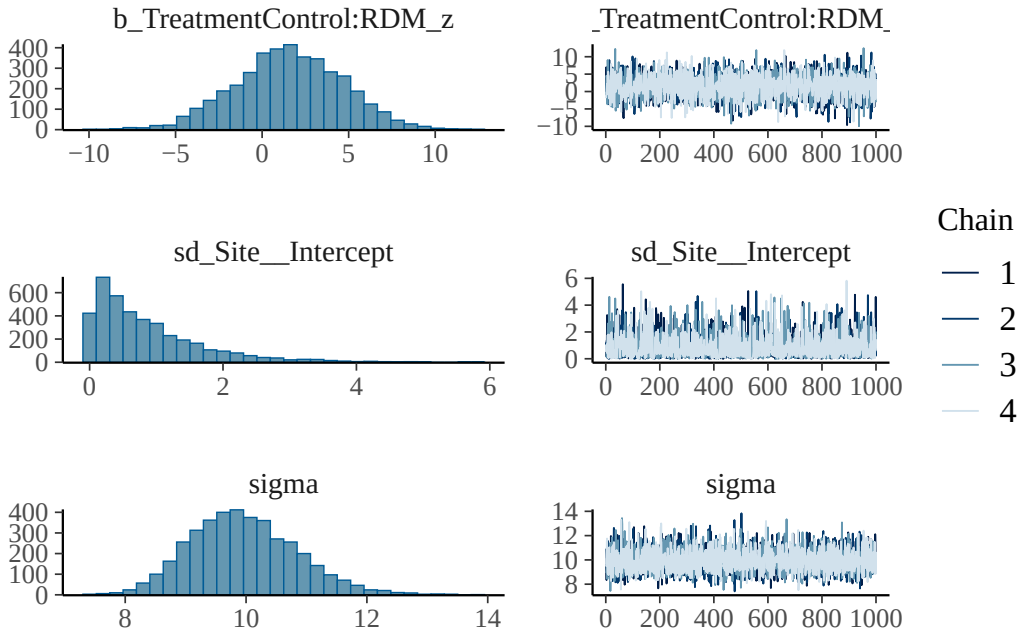
```r
m2_RDM <- brm(data = RDM_delta, # Reduced dataframe with burn-only transects
                family = gaussian, # normal distribution
                # Model with only severity and site random intercept
                final_RDM ~ Treatment * RDM_z + precip_z + clay_z + (1 | Site),
                # Specifying the standard priors
                prior = priors_RDM,
                # Defaults from class
                iter = 2000, warmup = 1000, chains = 4, cores = 4,
                # Change how cautious Stan explores posterior (default is 0.8)
                control = list(adapt_delta = 0.95),
                # Saving the model output
                file = "output/m.2_RDM")

# Check if the model is normal
summary(m2_RDM) #Rhat = 1
```

```
 Family: gaussian
  Links: mu = identity
Formula: final_RDM ~ Treatment * RDM_z + precip_z + clay_z + (1 | Site)
   Data: RDM_delta (Number of observations: 56)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~Site (Number of levels: 6)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     0.87      0.83     0.02     3.19 1.00     2554     1554

Regression Coefficients:
                       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
Intercept                -13.09      2.09   -17.14    -8.94 1.00     5040
TreatmentControl           8.50      2.53     3.53    13.45 1.00     5718
RDM_z                     -7.69      1.85   -11.23    -3.98 1.00     4636
precip_z                  -2.48      1.74    -5.97     0.92 1.00     5051
clay_z                     4.02      1.43     1.22     6.74 1.00     4678
TreatmentControl:RDM_z     1.55      3.13    -4.49     7.49 1.00     4632
                       Tail_ESS
Intercept                  3012
TreatmentControl           3160
```

```
RDM_z                       3064
precip_z                    3358
clay_z                      2753
TreatmentControl:RDM_z      3341

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     9.96      0.88     8.41    11.83 1.00     6180     3644

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
plot(m2_RDM)
```






```
# The posterior distributions are smooth with one clean peak, the four chains are overlapping
```

Now it's time to look at the best model fit for change in aboveground biomass. It looks like Model 1 (without the interaction) is a better fit with PSIS = 434.7 compared to Model 2 with 435.3. That is nice since Model 1 is easier to interpret.

```
# PSIS
loo(m0_RDM) #453.6
```

```
Computed from 4000 by 56 log-likelihood matrix.

         Estimate   SE
elpd_loo   -226.8  9.8
p_loo         5.7  2.0
looic       453.6 19.6
------
MCSE of elpd_loo is 0.1.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.5, 1.2]).

All Pareto k estimates are good (k < 0.7).
See help('pareto-k-diagnostic') for details.
```

```r
loo(m1_RDM) #434.7  --> best model fit (without the interaction)
```

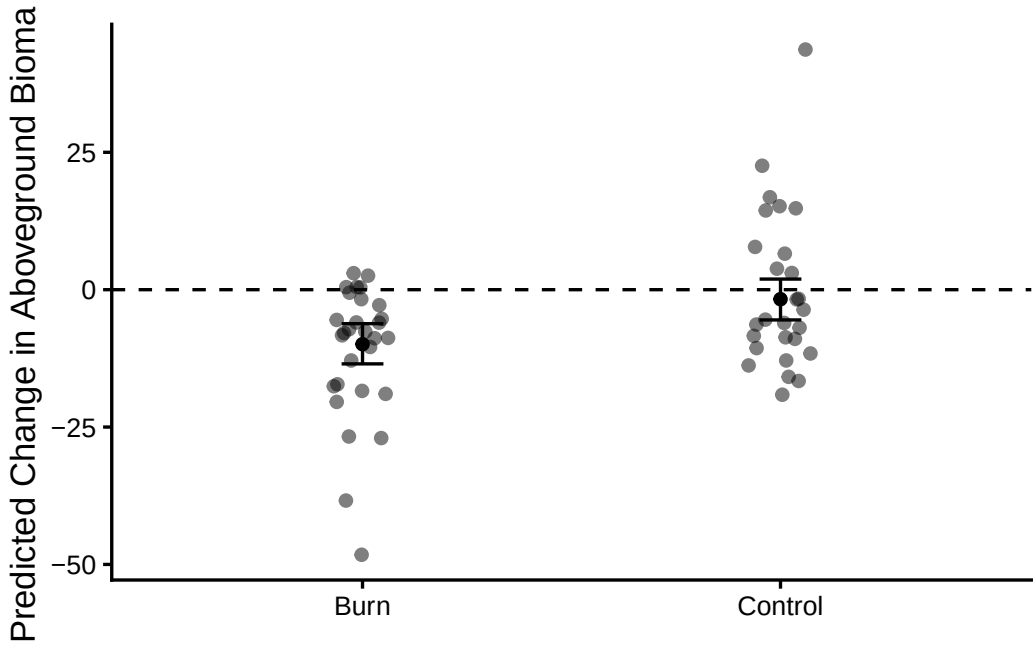
```
Computed from 4000 by 56 log-likelihood matrix.

         Estimate   SE
elpd_loo   -217.4 10.2
p_loo         8.4  3.0
looic       434.7 20.5
------
MCSE of elpd_loo is NA.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.7, 1.6]).

Pareto k diagnostic values:
                         Count Pct.    Min. ESS
(-Inf, 0.7]   (good)     55    98.2%   1050
   (0.7, 1]   (bad)       1     1.8%   <NA>
   (1, Inf)   (very bad)  0     0.0%   <NA>
See help('pareto-k-diagnostic') for details.
```

```r
loo(m2_RDM) #435.3
```

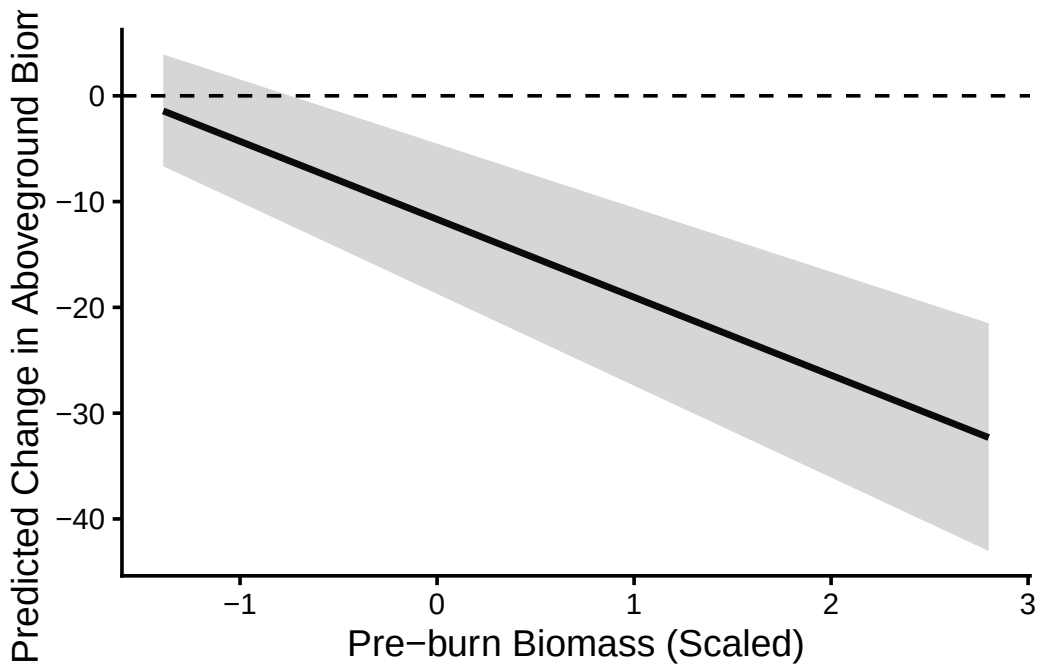
```
Computed from 4000 by 56 log-likelihood matrix.

         Estimate   SE
elpd_loo   -217.7 10.2
p_loo         8.9  3.2
looic       435.3 20.5
------
MCSE of elpd_loo is NA.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.6, 1.9]).

Pareto k diagnostic values:
                         Count Pct.    Min. ESS
(-Inf, 0.7]   (good)     55    98.2%   881
   (0.7, 1]   (bad)       1     1.8%   <NA>
   (1, Inf)   (very bad)  0     0.0%   <NA>
See help('pareto-k-diagnostic') for details.
```

Now, it's time to visualize the model outputs.

```r
# Load the libaries
library(ggeffects)
library(ggplot2)

# ΔRDM by treatment #
# Create the prediction
pred_treat_RDM <- ggpredict(m1_RDM, terms = "Treatment")

# Plot the prediction
rdm_pred<-
ggplot()+
  # Add the raw data
  geom_jitter(
    data = RDM_delta,
    aes(x = Treatment, y = final_RDM),
    width = 0.08,
    alpha = 0.5
  ) +
  # Add the mean point of the model prediction
  geom_point(
    data = pred_treat_RDM, aes(x = x, y = predicted)) +
  geom_point(size = 3) +
  # Add the standard error bar for model prediction mean
  geom_errorbar(
    data = pred_treat_RDM,
    aes(x = x, ymin = conf.low, ymax = conf.high), width = 0.1) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  theme_classic(base_size = 13) +
  labs(
    x = "",
    y = "Predicted Change in Aboveground Biomass)"
  )
plot(rdm_pred)
```

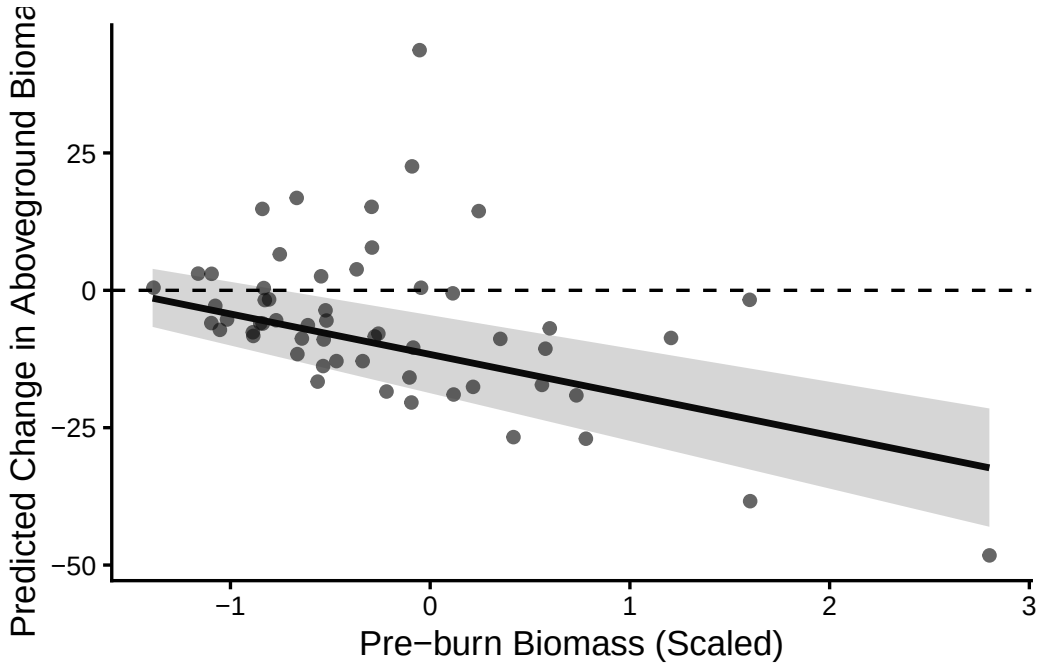



```
# Shows estimated mean change in RDM for Burn and Control from the model
# 95% credible intervals
# Easy to see that the Burn is more negative than Control
## The model predicted larger declines in aboveground biomass in burned transects than in co

# ΔRDM vs pre-burn biomass #
# Marginal effect of pre-burn biomass
# First, the prediction
pred_RDM <- ggpredict(m1_RDM, terms = "RDM_z [minmax]")

# Plot without raw values
ggplot(pred_RDM, aes(x = x, y = predicted)) +
  geom_line(linewidth = 1.2) +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.2) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  theme_classic(base_size = 14) +
  labs(
    x = "Pre-burn Biomass (Scaled)",
    y = "Predicted Change in Aboveground Biomass"
  )
```




```
# Plot with raw values
rdm_pred_raw <-
ggplot() +
  geom_point(
    data = RDM_delta,
    aes(x = RDM_z, y = final_RDM),
    alpha = 0.6
  ) +
  geom_line(
    data = pred_RDM,
    aes(x = x, y = predicted),
  linewidth = 1.2) +
  geom_ribbon(
    data = pred_RDM,
    aes(x = x, ymin = conf.low, ymax = conf.high), alpha = 0.2) +
  # Create a line at 0, so that decrease is below the line
  geom_hline(yintercept = 0, linetype = "dashed") +
  theme_classic(base_size = 13) +
  # Range of the dataset plotted
  coord_cartesian(xlim = range(RDM_delta$RDM_z))+
  labs(
    x = "Pre-burn Biomass (Scaled)",
    y = "Predicted Change in Aboveground Biomass"
```

```
  )
print(rdm_pred_raw)
```




```
## As pre-burn biomass increases, the predicted change in aboveground biomass (RDM) becomes
 # Higher fuel loads leads to greater biomass loss
 # Predicted biomass change declined with increasing pre-burn biomass, indication that trans

# Treat
```